HealthConnect Rwanda

Week 7 Project Summary, Data Analytics track

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#	Point	Summary
1	What I planned to test	The KPI calculations, the values shown in the dashboard, and whether the Week 6 conclusions hold across facilities, months, provinces and visit types.
2	What I actually tested	Thirty-two automated calculation tests recomputed from the raw schema, seven statistical robustness checks, and twenty-three dashboard checks against SQL on three successive exports (14 of the 20 initial checks passed, then 18 of 23, then 21 of 23).
3	Most important results	28 of 32 tests passed. Every financial total reproduced exactly. The 33% emergency share and the 90% collection rate are stable across segments. The satisfaction and revenue rankings are not.
4	Issues and weaknesses	Valid future dates rejected (864 of 5,000), outstanding amount ignoring balances on Paid bills (311 million), doctor workload split by department, ICU label casing, and in the dashboard a department matrix covering only 35% of values, a constant staff column, a payment visual showing billed instead of collected amounts, a decomposition tree 91% blank, a blocked map and colour rules that carry no information. Re-testing then caught a side effect and a regression caused by my own fixes (a filter that cut the diagnosis tree to 5,000 and a lost date link that flattened the complaint trend).
5	Improvements made	Date rule widened, outstanding computed as billed minus collected, workload grouped by doctor, source casing kept, dashboard relationships, measure and formats corrected, and rankings presented as descriptive.
6	Re-test results	Dates: 0 rejected. Outstanding: 919.9 million, matching billed minus collected. Dashboard: 21 of 23 checks pass in the final export, with the side effect and the regression both corrected. Still open: province map, readability of dark matrix cells and the ICU label. Query 14 was corrected but its output was not captured.
7	Tracks collaborated with	Data Science and Project Management (hand-over of validated findings and issues).
8	Tested collaboratively	Six validated facts for Data Science and eight issues for the Project Management log were prepared, and each fact is reproduced by a test. No joint test with another track was completed at the time of writing.
9	What changed as a result	Recommendations were adjusted so other tracks do not build on conclusions that failed testing. No response from the other tracks had been received at the time of writing.
10	Key findings	Collections are the clearest opportunity (920 million outstanding, 9.9%). One visit in three is an emergency everywhere. Facility rankings on satisfaction and revenue should not guide decisions yet.
11	Major challenges	Only 5,000 of 60,000 appointments exist, billing status contradicts payments, and the assignment referred to a different dataset.
12	Important decisions	Continue with the Week 6 dataset (error in the assignment confirmed), define outstanding from payments, withdraw the satisfaction-based recommendation, and present revenue rankings as descriptive.
13	Remaining limitations	No attendance status, admissions, laboratory, prescription or referral data, no facility names, and facility views covering about 8% of bills.
14	Remaining issues and dependencies	Open on my side: province map, readability of dark matrix cells, ICU label. Source defects (missing appointments, status and revenue-summary contradictions, department and facility mismatch) depend on the data owner.
15	Contribution to HealthConnect	Validated KPI definitions and totals, a documented list of data defects, and recommendations backed by tests.
16	Before Week 8	Fix the open dashboard items, apply the ICU label fix, record the responses of the other tracks, and re-test with the full appointment history if it becomes available.